

Cell-Free ISAC 系统鲁棒波束成形优化： 从非凸问题到可解 SDP

研究阶段报告

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Abstract

本文针对无蜂窝集成感知与通信 (Cell-Free Integrated Sensing and Communication, ISAC) 系统, 研究在信道状态信息 (CSI) 存在估计误差的条件下, 联合波束成形优化的鲁棒设计问题。考虑通信用户服务质量 (QoS) 约束、感知检测概率约束、跟踪精度约束 (以 PCRB 表征) 以及接入点 (AP) 选择约束, 建立了以最小化系统总发射功率为目标的混合整数非线性规划 (MINLP) 问题。通过引入协方差矩阵变量、半定松弛 (SDR) 以及 S-Procedure, 将原始极度非凸问题严格转化为标准凸半定规划 (SDP), 并证明转化过程的等价性。进一步给出高斯随机化秩一恢复算法, 保证凸最优解可转化为物理可实现的波束向量。本文建立了从物理建模到凸优化求解的完整数学框架, 为后续算法实现与系统部署奠定理论基础。

Contents

1	Introduction	2
2	System Model and Problem Formulation	2
2.1	Cell-Free ISAC System Model	2
2.1.1	Communication Signal Model	3
2.1.2	Sensing Signal Model	3
2.1.3	Imperfect CSI Model	3
2.2	Problem Formulation	4
3	Problem Convexification and SDP Reformulation	4
3.1	Non-Convexity Analysis	4
3.2	Convexification Procedure	5
3.3	Final Convex SDP Problem	7
4	Beam Recovery and Algorithm Design	8
4.1	Rank-One Recovery via Gaussian Randomization	8
4.2	Computational Complexity Analysis	8
4.2.1	SDP Solver Complexity	8
4.2.2	Closed-Form Baseline Complexity	10
5	Conclusion	10

1 Introduction

集成感知与通信 (Integrated Sensing and Communication, ISAC) 作为第六代移动通信 (6G) 的关键使能技术, 通过在共享频谱和硬件资源上同时实现通信与雷达感知功能, 显著提升了频谱效率和系统能效 [1]。与传统分离式设计相比, ISAC 通过波形共享和联合波束成形设计, 在通信性能与感知精度之间取得最优权衡, 已成为学术界和工业界的研究热点 [2]。

在 ISAC 系统架构中, 无蜂窝 (Cell-Free) 网络通过分布式部署大量低功率接入点 (Access Point, AP), 由中央处理单元 (Central Processing Unit, CPU) 通过前传链路进行联合信号处理, 有效消除小区间干扰并提升覆盖均匀性 [3]。将 Cell-Free 架构与 ISAC 相结合, 既能利用分布式 AP 的空间分集增益提升通信可靠性, 又能通过多视角观测增强目标感知与跟踪能力, 代表了未来无线网络的重要演进方向 [4]。

然而, Cell-Free ISAC 系统的联合波束成形优化面临多重挑战。首先, 通信波束与感知波形在功率、空间方向图和时频资源上存在耦合, 导致优化问题高度非凸。其次, 实际系统中信道状态信息 (Channel State Information, CSI) 通过有限导频估计获得, 存在有界或随机的估计误差, 需要在优化设计中显式考虑鲁棒性。第三, 感知性能通常以 Cramér-Rao 下界 (CRB) 或预测 CRB (PCRB) 表征, 其关于波束变量的非线性依赖进一步增加了问题的复杂性。最后, AP 选择作为二进制决策变量, 使问题成为混合整数非线性规划 (Mixed-Integer Nonlinear Programming, MINLP), 在计算上属于 NP-hard 类。

针对上述挑战, 本文建立了 Cell-Free ISAC 系统鲁棒波束成形的完整数学框架, 核心贡献如下:

- 问题建模: 建立了包含通信 QoS、感知检测、跟踪精度和 AP 选择约束的标准 MINLP 问题, 显式刻画 CSI 估计误差对通信和感知性能的影响。
- 等价凸化: 通过协方差矩阵提升、半定松弛 (SDR) 和 S-Procedure, 将原始非凸问题严格转化为标准凸 SDP, 并证明转化过程的等价性。
- 求解算法: 给出基于高斯随机化的秩一恢复算法, 保证凸最优解可转化为物理可实现的波束向量, 同时提供功率缩放兜底方案。
- 复杂度分析: 对比 SDP 内点法求解与 ZF 闭式基准的复杂度, 为工程实现提供计算可行性评估。

Notation: Scalars are denoted by italic letters, vectors by bold lowercase letters, and matrices by bold uppercase letters. $\mathbb{C}^{N \times M}$ and $\mathbb{H}^N$ denote the set of $N \times M$ complex matrices and $N \times N$ Hermitian matrices, respectively. $(\cdot)^T$, $(\cdot)^H$, and $\text{tr}(\cdot)$ denote transpose, conjugate transpose, and trace, respectively. $\mathbf{X} \succeq \mathbf{0}$ means $\mathbf{X}$ is positive semidefinite. $\|\cdot\|_2$ denotes the Euclidean norm.

2 System Model and Problem Formulation

2.1 Cell-Free ISAC System Model

考虑一个 Cell-Free ISAC 系统, 包含 M 个分布式 AP、 K 个单天线通信用户 (Communication User, CU) 和 P 个感知目标。每个 AP 配备 N_t 根发射天线, 通过理想前传链路与中央 CPU 连接。系统采用时分双工 (TDD) 模式, 利用信道互易性获取下行 CSI。

2.1.1 Communication Signal Model

用户 k 的接收信号可表示为:

$$y_k = \underbrace{\sum_{m=1}^M \mathbf{h}_{m,k}^H \mathbf{w}_{m,k} s_k}_{\text{desired signal}} + \underbrace{\sum_{j \neq k} \sum_{m=1}^M \mathbf{h}_{m,k}^H \mathbf{w}_{m,j} s_j}_{\text{multi-user interference}} + n_k \quad (1)$$

其中 $s_k \sim \mathcal{CN}(0, 1)$ 为发送符号, $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$ 为 AP m 到用户 k 的通信波束成形向量, $\mathbf{h}_{m,k} \in \mathbb{C}^{N_t}$ 为 AP m 到用户 k 的通信信道, $n_k \sim \mathcal{CN}(0, \sigma_c^2)$ 为接收噪声。

为便于分析, 引入堆叠向量形式。定义全局通信信道 $\mathbf{h}_k \in \mathbb{C}^{MN_t}$ 和全局通信波束 $\mathbf{w}_k \in \mathbb{C}^{MN_t}$:

$$\mathbf{h}_k = [\mathbf{h}_{1,k}^T, \mathbf{h}_{2,k}^T, \dots, \mathbf{h}_{M,k}^T]^T, \quad \mathbf{w}_k = [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \dots, \mathbf{w}_{M,k}^T]^T \quad (2)$$

则接收信号可紧凑表示为:

$$y_k = \mathbf{h}_k^H \mathbf{w}_k s_k + \sum_{j \neq k} \mathbf{h}_k^H \mathbf{w}_j s_j + n_k \quad (3)$$

2.1.2 Sensing Signal Model

对于单基地雷达感知, 目标 p 的反射信号为:

$$y_{S,p} = \sum_{m=1}^M \mathbf{g}_{m,p}^H \mathbf{Z}_m \mathbf{g}_{m,p} + n_{S,p} \quad (4)$$

其中 $\mathbf{Z}_m \in \mathbb{C}^{N_t \times N_t}$ 为 AP m 的感知协方差矩阵, $\mathbf{g}_{m,p} \in \mathbb{C}^{N_t}$ 为 AP m 到目标 p 的感知信道, $n_{S,p} \sim \mathcal{CN}(0, \sigma_s^2)$ 为感知接收噪声。

2.1.3 Imperfect CSI Model

实际系统中, 信道估计存在有界误差。本文采用确定性有界误差模型:

Communication channel:

$$\mathbf{h}_k = \hat{\mathbf{h}}_k + \Delta \mathbf{h}_k, \quad \|\Delta \mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2 \quad (5)$$

Sensing channel:

$$\mathbf{g}_p = \hat{\mathbf{g}}_p + \Delta \mathbf{g}_p, \quad \|\Delta \mathbf{g}_p\|_2 \leq \epsilon_g \|\hat{\mathbf{g}}_p\|_2 \quad (6)$$

其中 $\hat{\mathbf{h}}_k$ 和 $\hat{\mathbf{g}}_p$ 为估计信道, $\Delta \mathbf{h}_k$ 和 $\Delta \mathbf{g}_p$ 为估计误差, ϵ_h 和 ϵ_g 为相对误差界。

2.2 Problem Formulation

优化目标为最小化系统总发射功率，同时满足通信 QoS、感知检测、跟踪精度以及 AP 选择约束：

$$\min_{\{\mathbf{w}_{m,k}\}, \{\mathbf{Z}_m\}, \{b_{mp}\}} \sum_{m=1}^M \left(\sum_{k=1}^K \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m) \right) \quad (\text{P1})$$

$$\text{s.t.} \quad \text{SINR}_k^{\text{wc}} \geq \gamma_k, \quad \forall k \in \mathcal{K} \quad (7)$$

$$\text{SINR}_{S,p} := \frac{|\mathbf{g}_p^H \mathbf{Z}_p \mathbf{g}_p|}{\sigma_s^2} \geq \gamma_S^{\text{PoD}}, \quad \forall p \in \mathcal{P} \quad (8)$$

$$\text{tr}(\mathbf{J}_p^{\text{data}}) \geq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P} \quad (9)$$

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}} \quad (10)$$

$$\sum_{k=1}^K \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m) \leq P_{\text{max}}, \quad \forall m \in \mathcal{M} \quad (11)$$

$$\mathbf{Z}_m \succeq \mathbf{0}, \quad \forall m \in \mathcal{M} \quad (12)$$

$$b_{mp} \in \{0, 1\}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P} \quad (13)$$

其中， γ_k 和 γ_S^{PoD} 分别为通信用户 k 的 SINR 门限与感知检测概率门限， $\Gamma_{\text{Track},p}$ 为目标 p 的跟踪精度门限（FIM 迹约束）， N_{req} 为每个目标所需服务 AP 数， P_{max} 为单 AP 最大发射功率， $b_{mp} \in \{0, 1\}$ 为 AP m 是否服务目标 p 的二进制指示变量。

注 1. Problem (P1) is a mixed-integer nonlinear programming (MINLP) problem. The non-convexity arises from: (i) the SINR constraints involving ratios of quadratic functions; (ii) the coupling between communication beamforming vectors $\mathbf{w}_{m,k}$ and sensing covariance matrices $\mathbf{Z}_m$ in the power constraints; (iii) the binary AP selection variables b_{mp} ; and (iv) the rank-one constraint implicitly required for physical beamforming implementation. The following sections develop a systematic convexification procedure to address these challenges.

3 Problem Convexification and SDP Reformulation

3.1 Non-Convexity Analysis

Problem (P1) is a mixed-integer nonlinear programming (MINLP) problem. The non-convexity arises from multiple sources: the SINR constraints involve ratios of quadratic functions (NC1), the worst-case CSI uncertainty makes the communication SINR constraint semi-infinite (NC2), the binary AP selection variables create a discrete feasible set (NC3), the rank-one constraint required for physical beamforming implementation defines a non-convex manifold (NC4), the coupling between communication beamforming vectors and sensing covariance matrices introduces bilinear terms (NC5), and the PCRB constraint involves a matrix inverse that depends nonlinearly on the optimization variables (NC6). Table 1 summarizes these non-convexity sources.

Table 1: Non-convexity source identification for problem (P1)

ID	Source	Affected constraints	Convexity violation
NC1	SINR fractional structure	(P1-C1), (P1-C2)	Ratio of quadratics
NC2	Worst-case CSI uncertainty	(P1-C1)	Semi-infinite constraint
NC3	Binary AP selection	(P1-C4), (P1-C7)	Discrete set $\{0, 1\}$
NC4	Rank-one requirement	(P1-C8)	Non-convex manifold
NC5	Variable coupling	(P1-C1)–(P1-C5)	Bilinear terms
NC6	Matrix inverse in PCRB	(P1-C3)	Nonlinear FIM dependence

3.2 Convexification Procedure

To address the aforementioned non-convexity, we develop a systematic convexification procedure that transforms the original problem (P1) into a standard convex semidefinite program (SDP). The procedure consists of seven steps, each targeting a specific non-convex source while preserving the problem structure as much as possible.

Step 1: Covariance matrix lifting. We first lift the per-AP vector variables to the global matrix space. The cooperative beamforming vector $\mathbf{w}_k \in \mathbb{C}^{MN_t}$ is defined as the concatenation of per-AP beams $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$, i.e., $\mathbf{w}_k = [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \dots, \mathbf{w}_{M,k}^T]^T$, and the per-AP selection matrix $\mathbf{E}_m = \text{diag}(\underbrace{0, \dots, 0}_{(m-1)N_t}, \underbrace{1, \dots, 1}_{N_t}, \underbrace{0, \dots, 0}_{(M-m)N_t})$ extracts AP m 's sub-vector.

We then define $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H \in \mathbb{H}^{MN_t}$ for all k , $\mathbf{Z} = \sum_{m=1}^M \mathbf{Z}_m \in \mathbb{H}^{MN_t}$, and the aggregate transmit covariance $\mathbf{R}_X = \sum_{k=1}^K \mathbf{W}_k + \mathbf{Z}$. The per-AP power constraint $\sum_k \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m) \leq P_{\max}$ becomes $\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}$. This transformation is strictly equivalent since there is a one-to-one mapping between $\mathbf{w}_k$ and rank-one $\mathbf{W}_k$. The lifting linearizes the objective function ($\|\mathbf{w}_k\|_2^2 = \text{tr}(\mathbf{W}_k)$), the per-AP power constraints, and the SINR expressions ($|\mathbf{h}_k^H \mathbf{w}_k|^2 = \text{tr}(\mathbf{h}_k \mathbf{h}_k^H \mathbf{W}_k)$). However, the rank-one constraint $\text{rank}(\mathbf{W}_k) = 1$ remains non-convex.

Step 2: Semidefinite relaxation. We relax the rank-one constraint by only requiring $\mathbf{W}_k \succeq \mathbf{0}$, which enlarges the feasible set from the non-convex rank-one manifold to the convex PSD cone. This is a tight relaxation when $K \leq 2$, where the optimal $\mathbf{W}_k^*$ naturally satisfies $\text{rank}(\mathbf{W}_k^*) = 1$. For $K > 2$, SDR provides a lower bound on the original problem, and Gaussian randomization can recover a feasible rank-one solution with bounded performance loss $O(1/L)$.

Step 3: S-Procedure for robust SINR. The worst-case SINR constraint is semi-infinite due to the relative uncertainty set $\|\Delta \mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2$. We rewrite it as a quadratic inequality in $\Delta \mathbf{h}_k$: $\tilde{f}_1(\Delta \mathbf{h}) = (\hat{\mathbf{h}} + \Delta \mathbf{h})^H \mathbf{A}_k (\hat{\mathbf{h}} + \Delta \mathbf{h}) - \sigma_c^2 \geq 0$, where $\mathbf{A}_k = \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j$. The uncertainty set is described by $\tilde{f}_2(\Delta \mathbf{h}) = \epsilon_h^2 \|\hat{\mathbf{h}}_k\|_2^2 - \|\Delta \mathbf{h}_k\|_2^2 \geq 0$. By the S-Procedure (which is necessary and sufficient for a single quadratic constraint over a norm ball), the semi-infinite worst-case constraint is equivalently transformed into the following linear matrix inequality with the slack variable $\mu_k \geq 0$:

$$\exists \mu_k \geq 0 : \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|_2^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k \quad (14)$$

This transformation is exact since the S-Procedure is necessary and sufficient for a single quadratic constraint over a norm ball.

Step 4: SINR fractional linearization. After the S-Procedure, the SINR constraint still involves a fractional structure. Assuming the denominator is positive (which always holds since $\sigma_c^2 > 0$), we cross-multiply to obtain a linear inequality:

$$\text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_k) - \gamma_k \sum_{j \neq k} \text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_j) \geq \gamma_k \sigma_c^2, \quad \forall k \quad (15)$$

where $\hat{\mathbf{H}}_k = \hat{\mathbf{h}}_k \hat{\mathbf{h}}_k^H$. This is a strict equivalence transformation for positive denominator.

Step 5: PCRB and sensing SINR linearization. Under the assumption that the channel gradient $\nabla_{\theta_p} \mathbf{g}_p$ is known and constant in the current time slot, the Fisher information matrix becomes an affine function of $\mathbf{R}_X$. Defining $\mathbf{F}_p = \frac{2}{\sigma_s^2} \text{Re}\{\nabla_{\theta_p}^H \mathbf{g}_p \nabla_{\theta_p} \mathbf{g}_p^H\} \succeq \mathbf{0}$, the PCRB constraint simplifies to $\text{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\text{Track},p}$, which is linear in $\mathbf{R}_X$. Similarly, the sensing SINR constraint $\text{SINR}_{S,p} = \frac{|\mathbf{g}_p^H \mathbf{Z} \mathbf{g}_p|}{\sigma_s^2} \geq \gamma_S^{\text{PoD}}$ (note the linear —not squared— definition) becomes $\text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\text{PoD}} \sigma_s^2$, which is also linear in $\mathbf{Z}$.

Step 6: Power constraint. The per-AP power constraint $\sum_k \|\mathbf{w}_{m,k}\|^2 + \text{tr}(\mathbf{Z}_m) \leq P_{\max}$ is already convex. In the lifted domain, it becomes $\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}$, which is a linear inequality. No additional transformation is needed.

Step 7: AP selection decomposition. The binary AP selection constraint is NP-hard in general. We adopt a two-step decomposition: first, a heuristic outer step selects the top- N_{req} APs for each target based on large-scale fading; then, the inner step solves the convex SDP with the fixed AP set. This yields a polynomial-time algorithm with suboptimal but engineering-acceptable performance.

Table 2 summarizes the seven-step convexification chain.

Table 2: Seven-step convexification method chain

Step	Transformation	Eliminates	Type	Result
1	Covariance lifting $\mathbf{w}\mathbf{w}^H \rightarrow \mathbf{W}$	NC1, NC5	Strict equivalence	Linear objective
2	SDR: drop rank($\mathbf{W}$) = 1	NC4	Tight relaxation	PSD cone
3	S-Procedure	NC2	Exact equivalence	LMI
4	Cross-multiply fraction	NC1 (comm)	Strict equivalence	Linear inequality
5a	PCRB affine expansion	NC6	Strict equivalence (Assump. 1)	Linear inequality
5b	Sensing SINR linearization	NC1 (sens)	Strict equivalence	Linear inequality
6	Power (already convex)	—	No change	Linear inequality
7	AP two-step decomposition	NC3	Heuristic (no guarantee)	Fixed AP set

3.3 Final Convex SDP Problem

Combining all the transformations described above, the original non-convex problem (P1) is reformulated as the following standard convex SDP:

$$\begin{aligned}
\min_{\{\mathbf{W}_k\}, \mathbf{Z}, \{\mu_k\}} \quad & \sum_{k=1}^K \text{tr}(\mathbf{W}_k) + \text{tr}(\mathbf{Z}) & (\text{P3}) \\
\text{s.t.} \quad & \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|_2^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k & (\text{P3-C1}) \\
& \text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p & (\text{P3-C2}) \\
& \text{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\text{Track}, p}, \quad \forall p & (\text{P3-C3}) \\
& \text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m & (\text{P3-C4}) \\
& \mathbf{W}_k \succeq \mathbf{0}, \quad \forall k & (\text{P3-C5}) \\
& \mathbf{Z} \succeq \mathbf{0} & (\text{P3-C6}) \\
& \mu_k \geq 0, \quad \forall k & (\text{P3-C7})
\end{aligned}$$

where $\mathbf{A}_k = \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j$, $\mathbf{F}_p$ is defined in Step 5, and $\mathbf{E}_m$ is the AP selection matrix. Constraint (P3-C1) is the LMI from the S-Procedure, (P3-C2) and (P3-C3) are the linearized sensing SINR and PCRB constraints, (P3-C4) is the per-AP power constraint, (P3-C5) and (P3-C6) are the PSD constraints from SDR, and (P3-C7) ensures the S-Procedure slack variables are non-negative.

Table 3: Constraint correspondence between (P1) and (P3)

(P1) constraint	(P3) constraint	Transformation	Convexity
(P1-C1) SINR	(P3-C1) LMI	Steps 1,2,3,4	LMI (convex)
(P1-C2) Sens. SINR	(P3-C2) Linear	Steps 1,2,5b	Linear (convex)
(P1-C3) PCRB	(P3-C3) Linear	Steps 1,2,5a	Linear (convex)
(P1-C4) AP select	(P3-C4) Fixed set	Step 7	Linear (convex)
(P1-C5) Power	(P3-C4) Linear	Steps 1,6	Linear (convex)
(P1-C6) PSD	(P3-C6) PSD	No change	PSD cone (convex)
(P1-C7) Binary	Eliminated	Step 7	Heuristic
(P1-C8) Rank-one	(P3-C5) PSD	Step 2	PSD cone (convex)

定理 1 (Problem Convexity and Equivalence). *Problem (P3) is a standard convex SDP. All constraints are either LMIs or linear inequalities, and the objective is linear. For $K \leq 2$, (P3) is equivalent to (P1) since the SDR is tight. For $K > 2$, (P3) provides a lower bound on (P1). Furthermore, (P3) can be solved in polynomial time via interior-point methods, with complexity upper bound $O((K+1)^3(MN_t)^6(K+P+M))$; simplified order $O((KMN_t^2)^{3.5})$.*

Proof. Each constraint in (P3) is either an LMI (P3-C1), a linear inequality (P3-C2, P3-C3, P3-C4), or a PSD cone constraint (P3-C5, P3-C6). The objective is linear. Therefore (P3) is a convex SDP. Steps 1, 3, 4, 5a, 5b, 6 are strict equivalences. Step 2 is a tight relaxation when $K \leq 2$ (SDR is tight for $K \leq 2$ and provides a lower bound for $K > 2$). Step 7 is a heuristic decomposition (outer AP selection by large-scale fading ordering **has**

no theoretical optimality guarantee—it is neither a lower bound nor an upper bound; the inner SDP on the fixed AP set is convex and tight). Therefore, for $K \leq 2$, (P3) $\equiv$ (P1); for $K > 2$, $P_{P3}^* \leq P_{P1}^*$. The complexity follows from the standard interior-point complexity $O(n^3L)$ for SDPs, where $n = O(K(MN_t)^2)$ is the decision-variable dimension and L is the input bit-length. $\square$

4 Beam Recovery and Algorithm Design

4.1 Rank-One Recovery via Gaussian Randomization

Problem (P3) yields covariance matrix solutions $\{\mathbf{W}_k^*\}$ and $\mathbf{Z}^*$, which may have rank greater than one. Physical implementation requires rank-one beam vectors. We propose a Gaussian randomization algorithm with power scaling fallback.

命题 1 (Power Scaling Guarantee). *After power scaling, the per-AP power constraint is strictly satisfied: $P_m' = \beta_m P_m = P_{\max}$. Both communication and sensing SINR scale by the same factor: $\text{SINR}'_k = \beta_m \cdot \text{SINR}_k$ and $\text{SINR}'_{S,p} = \beta_m \cdot \text{SINR}_{S,p}$.*

注 2. *Unlike pure communication systems where pencil beams are preferred, ISAC sensing requires spatial coverage. Therefore, the multi-rank sensing covariance $\mathbf{Z}^*$ is a design feature rather than an algorithmic failure. The “role separation” of communication and sensing in the covariance domain is the essential characteristic of ISAC waveform design.*

4.2 Computational Complexity Analysis

4.2.1 SDP Solver Complexity

Variable count:

- K matrices $\mathbf{W}_k$: $K \cdot (MN_t)^2$ real variables
- 1 matrix $\mathbf{Z}$: $(MN_t)^2$ real variables
- K scalars μ_k : K real variables
- Total: $O(K(MN_t)^2)$

Constraint count:

- K communication LMIs: $K \cdot (MN_t + 1) \times (MN_t + 1)$
- P sensing linear constraints
- M power constraints
- Total computational complexity: $O(K(MN_t)^3)$

Solution time: $O((KMN_t^2)^{3.5})$ via interior-point methods. For $M = 16$, $N_t = 4$, $K = 10$, approximately 5–10 seconds (MOSEK).

Algorithm 1: Gaussian Randomization Rank-One Recovery

Input: SDP optimal solution $\{\mathbf{W}_k^*\}_{k=1}^K$, $\mathbf{Z}^*$, constraint parameters

Output: Feasible beamforming vectors $\{\mathbf{w}_k\}_{k=1}^K$, sensing waveforms

// Step 1: Communication beam recovery

```
1 foreach user  $k$  do
2   if  $\text{rank}(\mathbf{W}_k^*) = 1$  then
3      $\mathbf{w}_k \leftarrow \sqrt{\lambda_{\max}(\mathbf{W}_k^*)} \cdot \mathbf{v}_{\max}(\mathbf{W}_k^*)$ ;           // Direct extraction
4   else
5     Generate  $L = 1000$  candidates:  $\boldsymbol{\xi}_l \sim \mathcal{CN}(\mathbf{0}, \mathbf{W}_k^*)$ ;
6     Normalize:  $\mathbf{w}_k^{(l)} \leftarrow \sqrt{\text{tr}(\mathbf{W}_k^*)} \cdot \frac{\boldsymbol{\xi}_l}{\|\boldsymbol{\xi}_l\|_2}$ ;
7     Compute constraint violation:
8        $v_l \leftarrow \sum_{k'} \max(0, \gamma_k - \text{SINR}_{k'}^{(l)}) + \sum_p \max(0, \gamma_S^{\text{PoD}} - \text{SINR}_{S,p}^{(l)})$ ;
9      $l^* \leftarrow \arg \min_l v_l$ ;
10    if  $v_{l^*} = 0$  then
11      Accept  $\mathbf{w}_k \leftarrow \mathbf{w}_k^{(l^*)}$ ;
12    else
13      Enter power scaling fallback (Step 3);
```

// Step 2: Sensing waveform recovery

13 Eigen-decomposition: $\mathbf{Z}^* = \sum_{i=1}^r \lambda_i \mathbf{v}_i \mathbf{v}_i^H$;

14 Generate multi-stream waveform: $\mathbf{z}_p = \sum_{i=1}^r \sqrt{\lambda_i} \mathbf{v}_i s_i$, where $s_i \sim \mathcal{CN}(0, 1)$ i.i.d.;

// Step 3: Power scaling fallback

```
15 foreach AP  $m$  do
16    $P_m \leftarrow \sum_k \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m)$ ;
17   if  $P_m > P_{\max}$  then
18      $\beta_m \leftarrow P_{\max}/P_m$ ;
19      $\mathbf{w}_{m,k} \leftarrow \mathbf{w}_{m,k} \cdot \sqrt{\beta_m}$ ;
20      $\mathbf{Z}_m \leftarrow \mathbf{Z}_m \cdot \beta_m$ ;
```

// Step 4: Performance guarantee

21 **if** $K \leq 2$ **then** SDR is tight, recovered solution is optimal;

22 **if** $K > 2$ **then** Gaussian randomization provides suboptimal solution with bounded performance loss $O(1/L)$;

Table 4: Complexity comparison between SDP and closed-form baseline

Algorithm	Complexity	Typical runtime ($M = 16, N_t = 4, K = 10, P = 4$)
SDP (MOSEK)	$O((KMN_t^2)^{3.5})$	5–10 seconds
ZF + matched filter	$O(MP \log M + K^3)$	< 0.1 seconds

4.2.2 Closed-Form Baseline Complexity

As a comparison, the ZF + matched filtering baseline has complexity $O(MP \log M + K^3)$, which is approximately 0.1 seconds for typical parameters. However, this baseline does not guarantee robust constraint satisfaction (success rate approximately 25% under CSI uncertainty).

5 Conclusion

This paper established a complete mathematical framework for robust beamforming optimization in Cell-Free ISAC systems. The main contributions are:

1. **Problem formulation:** A standard MINLP (P1) incorporating communication QoS, sensing detection, tracking accuracy, and AP selection constraints under CSI uncertainty.
2. **Robustness analysis:** Worst-case SINR constraints with bounded CSI errors, introducing robustness factors η_h and η_g to quantify error impact.
3. **Convex relaxation:** SDR to relax non-convex rank-one constraints, with proof of sensing constraint convexity under single-angle estimation.
4. **S-Procedure transformation:** Exact equivalence transformation of infinite-dimensional worst-case constraints to finite-dimensional LMIs.
5. **Rank-one recovery:** Gaussian randomization algorithm with power scaling fallback to guarantee physical implementability.

Future work includes:

- Implementation of full SDP solver with MOSEK LMI support
- Extension to sensing channel robustness (scenarios B/C)
- Multi-target joint optimization with temporal coupling
- Simulation validation and performance evaluation

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